

Materials and Methods

Data Preprocessing and Unit Selection

Spatial firing rates during cued trials were extracted for each animal across a 500 cm virtual linear track, discretised into 200 spatial bins (2.5 cm/bin). Neural activity was analysed across six target regions of interest: CA1, CA3, Dentate Gyrus (DG), Subiculum (SUB), Primary Visual Cortex (V1), and Retrosplenial Cortex (RSC). To isolate putative pyramidal cell activity, fast-spiking (FS) interneurons were systematically excluded from the V1, RSC, CA1, and CA3 datasets based on their waveform characteristics. A minimum inclusion criterion of 5 isolated, non-FS units per region per animal was applied; regions failing to meet this threshold in a given animal were excluded from pairwise analyses.

Dimensionality Reduction (Principal Component Analysis)

To extract dominant population-level activity patterns and standardise the dimensionality across regions with varying unit yields, we applied Principal Component Analysis (PCA) to the trial-concatenated spatial firing matrices. For each region, the 3D data tensor (Units $\times$ Spatial Bins $\times$ Trials) was flattened into a 2D matrix ((Spatial Bins $\times$ Trials) $\times$ Units). PCA was applied to this concatenated matrix, and the resulting principal component scores were subsequently reshaped back into a 3D structure (Components $\times$ Spatial Bins $\times$ Trials) for downstream analysis. We dynamically selected the number of principal components (PCs) to retain based on a cumulative variance threshold, keeping the minimum number of components required to explain 90% of the total variance.

Shared Variance Analysis (Canonical Correlation Analysis)

To quantify the inter-regional shared variance, we applied Canonical Correlation Analysis (CCA) to the PCA-reduced population trajectories. Eight directed anatomical pairs were analysed: CA1-V1, CA1-DG, CA1-CA3, CA1-RSC, CA1-SUB, V1-RSC, RSC-SUB, and CA3-DG. For each pair, CCA was computed to find linear combinations of the regional PCs that maximally correlated with one another. The magnitude of shared variance was defined by the first canonical correlation coefficient (CC1) returned by MATLAB’s *canoncorr* function.

To capture the spatiotemporal dynamics of this coupling, CCA was executed in two parallel streams:

1. **Trial-wise CCA:** Computed using a 7-trial sliding window (centred on trial t , ± 3 trials) to capture trial-by-trial evolution of inter-regional coupling.
2. **Bin-wise CCA:** Computed using a 7-bin sliding window (centred on bin b , ± 3 bins) across the spatial extent of the track to identify spatially localized peaks in inter-regional coordination.

To establish chance-level baselines, we generated null distributions for both streams using 50 iterations of randomised shuffling. For trial-wise CCA, the spatial bins within the sliding window were independently permuted. For bin-wise CCA, the trial order within the spatial window was independently permuted.

Dual-Region Precession Index

To determine the directional lead-lag relationship (precession) between coupled brain regions, we cross-correlated the first canonical variates of each pair across temporal/spatial shifts ranging from -2 to $+2$ lags. We defined a normalised Information Flow Index (IFI) as:

$$IFI = \frac{\bar{r}_{\text{negative_lags}} - \bar{r}_{\text{positive_lags}}}{\bar{r}_{\text{negative_lags}} + \bar{r}_{\text{positive_lags}}} \quad (1)$$

where $\bar{r}_{\text{negative_lags}}$ is the mean canonical correlation when Region 1 leads Region 2, and $\bar{r}_{\text{positive_lags}}$ is the mean correlation when Region 2 leads Region 1. The index is bounded between -1 and 1 . A positive index indicates that Region 1’s activity reliably precedes Region 2, whereas a negative index indicates Region 2 precedes Region 1. If the sum of the correlations was negligible (< 0.001), the index was excluded to prevent noise-driven amplification.

Behavioural Epochs and Non-learner Controls

To assess learning-dependent network reorganization, animals were segregated into Learners and Non-Learners based on behavioural performance. For Learners, a discrete learning point (LP) was identified as described above. To control for total exposure time, each Non-Learner was assigned a pseudo-LP equal to the rounded mean LP of the Learner cohort (51 trials). Data were then aligned to these LPs and binned into three 10-trial epochs:

- **Early (Naive):** Trials 1 to 10.
- **Pre-Learning (Intermediate):** 10 trials immediately preceding the LP.
- **Post-Learning (Expert):** 10 trials starting exactly at the LP.

Statistical Analysis and Network Visualization

Group-level differences across epochs and between learning states were evaluated using a Mixed Repeated Measures Analysis of Variance (rmANOVA). “Epoch” (Early, Pre, Post) served as the within-subjects repeated measure, and “Group” (Learner vs. Non-Learner) served as the between-subjects factor. Analyses were conducted on animal-averaged means for both CC1 and the Information Flow Index. Post-hoc pairwise comparisons were conducted where main effects or interactions were significant.

Additionally, we assessed whether inter-regional directional leads were statistically significant at a given epoch using 1-sample t-tests comparing the animal-averaged Precession Index against a null hypothesis of 0 (no directional lead).

To visualize network states, we constructed network graphs for the Naive and Expert epochs. Edge thickness represents the average magnitude of functional coupling (CC1) during a given epoch, while edge direction and colour intensity depict the magnitude and direction of the Information Flow Index. Significant directional coupling (derived from the 1-sample t-tests against zero) was denoted with asterisks on the respective edges.